GFZ Helmholtz Centre for Geosciences

(*Pre-print*)

Rheolopy: A python tool for analyzing yield strength parameters

Abstract

To understand the strength distribution in the Earth's rigid outermost shell, i.e, the lithologically heterogeneous lithosphere, extensive knowledge about the stress-strain (rate) relations, given by constitutive laws, of various rock types under variable pressure and temperature conditions is required. Therefore, it is important, how different rocks deform under stress. This deformation is described by constitutive laws, describing the material behaviour under a given stress, which are dependent on the rheological properties of the rocks, which are defined by their mineralogical compositions as well as the temperature and pressure conditions they are exposed to. The relationship between stress and deformation in rocks are derived from laboratory experiments under controlled conditions. The resulting formulas contain a diverse set of coefficients, which are not well constrained and sometimes state sensitive. Furthermore they are usually derived from single minerals and thus not representative of an aggregate of minerals as it occurs in nature, which makes the constitutive laws inconsistent across literature. This poses a challenge for geophysical modeling, where rheological parameters must be consistent within the model to ensure that variations arising from different measurement setups do not introduce inconsistencies into the results. To facilitate the decision-making process in the choice of coefficients and enable a comparison of rheological parameters for different materials (rock types and minerals), the rheolopy package has been developed.

Keywords: rheology; rock properties; lithospheric strength; lithology;

1. Introduction

Rheology describes the mechanical behaviour of rocks in response to applied stress. The nature of this response is highly dependent on the physical conditions prevailing during the deformation, such as temperature, pressure, strain rate or fluids present as well as on rock composition. The deformation itself can either be reversible (recoverable) or irreversible (permanent). While reversible deformation, which includes elastic deformation, is described by a time-independent, often linear relationship between stress and strain, irreversible deformation is characterized by a nonlinear stress-strain (rate) relationship. To mathematically describe the relation between the state of stress applied to a rock volume and its deformation or deformation rate, constituive laws are used. These laws are derived from laboratory experiments, where rock samples are subjected to forces and stresses under controlled conditions and their deformation is measured by a rock deforming apparatus. Based on an Arrhenius law, the equations are emperically fit to match the observations. The constitutive equations involve both intrinsic material properties and extrinsic external conditions, such as temperature, pressure, strain rate or fluids present. In general, different rock types will respond differently to the same level of stress, yet some general observations can be made. For example, rocks tend to deform either elastically or by brittle failure at low confining pressures and temperatures and over a short time scale, as is the case at shallow crustal levels (<15 km depth). High confining pressures and temperatures lead to ductile deformation, where the rock deforms steadily, upon a tipping point, without loosing its bulk coherency. It should be noted, that even though the effect of creep is usually minimal and negligible for upper crustal conditions, it still does occur in these regimes.

1.1 Brittle Deformation

For brittle behaviour, rock failure occurs when the applied stress exceeds the static strength of the rock, where it ruptures and losses its bulk coherence. Deformation at the yield point is a type of localized (with an additional damage zone), permanent, plastic deformation characterized by loss of cohesion by fracturing (nucleation or coalescence). The level of localization usually scales with the level of confinement. This is only valid for intact rocks or newly forming fracture planes. After that brittle failure occurs in the form of preferential deformation along weak fracture planes. Ductile behaviour on the other hand represents the continous redistribution of permanent strain.
Hence it is important for geophysical modeling to assess the ability of rocks to deform under certain pressure and temperature conditions. Thus creep is essentially cast as an exponential function of the background temperature. The rock's strength can be described by the differential stress ($\Delta\sigma$), at which the rock starts to experience brittle or ductile deformation:

$$\Delta\sigma = \sigma_1 - \sigma_3 \tag{1}$$

where σ_1 is the maximum principal stress and σ_3 is the minimum principal stress.
At shallow depth, rocks predominantly deform by brittle behaviour. This is characterized by a linear relationship between the shear stress τ and the normal stress σ_N along a fracture or fault plane, which is approximated via a linearized Mohr-Coulomb failure criterion (Jaeger, Cook, and Zimmerman 2007):

$$\tau = \mu(\sigma_N - P_f) + c \tag{2}$$

where c is the cohesion (shear strength), μ is the friction coefficient and σ_N is the total normal stress, and P_f is the pore fluid pressure, so that $\sigma_N - P_f$ represents the effective normal stress. For consolidated rocks, the cohesion is often assumed to be 0, leading to the simplified form of the Mohr-Coulomb failure criterion:

$$\tau = \mu(\sigma_N - P_f) \tag{3}$$

Byerlee's brittle failure (Byerlee 1978), which describes the amount of shear stress required for failure for different rock types, comes upon equalling Mohr-Coulomb failure with the shear strength of a rock. It is often used to estimate the friction coefficient μ for different rock types, which can be used to assess the rocks ability to deform. Especially in the upper crust, where discontinuities such as faults and fractures are common in every shape and size, the relevant failure processes take place at these pre-existing planes in the form of sliding. Sibson (1974) therefore proposed a modification which quantifies the linear failure criterion in terms of differential stress, while also accounting for the effects of pore fluid pressure and overburden pressure (Sibson 1974):

$$\Delta\sigma_B = \beta\rho_{bulk} g z(1 - \lambda) \tag{4}$$

where ρ_{bulk} is the bulk density, g is the gravitational acceleration, z is the depth and λ is the hydrostatic pore fluid factor. Given its dependancy on the applied confining pressure, it is also a function of the fluid pressure within the rock. The fluid within the pore space of the rock reduces the normal stress component on existing planes within the porous medium, so that failure may occure at lower differential stresses. It should be noted that Eq. 4 represents Eq. 3 cast in terms of principal differential stresses, assuming an Andersonian stress state where one principal stress is vertical (representing the overburden pressure). For typical applications in geodynamics, one can assume eq. 4 to be valid, even though eq. 3 should be be considered once the normal stress is is the effective one. Furthermore, Sibson's approximation is also dependent on the stress regime of the surrounding rock, which can either be compressional or extensional (thrust or normal faulting). This stress regime should be Andersonian. He demonstrated that the minimum values of $R' = \sigma_1/\sigma_3$ required to initiate sliding on thrust, strike-slip and normal faults, respectively, take relative differences corresponding to the ratios [4 : 1.6 : 1]. The magnitude of differential stress varies significantly depending on the prevailing stress regime. Compressional stress regimes are characterized by a horizontal σ_1 (maximum principal stress) and a vertical σ_3 (minimum principal stress), and are commonly associated with thrust faulting, where faults typically dip at an angle of 30°. In contrast, tensional stress regimes are defined by a vertical σ_1 and a horizontal σ_3, leading to the development of normal faults that tend to dip at an angle of 60°. Considering a thrust fault and a normal fault at the same depth, therefore experiencing the same load (representing overburden pressure), the minimum principal stress (σ_3) in the compressional regime equals this load, while the maximum principal stress (σ_1) equals the load in the tensional regime. To overcome the normal stress and initiate slip on a fault, a critical differential stress is required. In a compressional stress regime, this critical differential stress for compression ($\Delta\sigma_C = \sigma_1 - load$) is approximately four times larger than the critical differential stress for extension ($\Delta\sigma_E = load - \sigma_3$) in a tensional stress regime. Therefore, $\beta_C = 3$ and $\beta_E = 0.75$ for compressional and extensional regimes respectively.

Ductile behaviour, together with brittle behaviour, represents one of the two primary deformation mechanisms in rocks. Creep defines the time-dependent flow of solid matter under constant stress, which can either be viscous or plastic flow. Its primary driver is temperature, so it prevails in deeper crustal or lithospherice levels or regions of high thermal gradients. Other important parameters for controlling creep are stress magnitude, strain rate and mineral composition. Similar to frictional materials, creep analysis by laboratory experiments often only yields quantitative constraints on their flow laws. Deformation by creep under these conditions can be partitioned into different regimes, which are characterized by different stress and strain relationships. The first regime is primary (transient / decelerating) creep, where the strain rate decreases progressivly in a logarithmic manner as a function of time. Secondary (steady state) creep is characterized by a constant strain rate, which is independent of time. For this reason, it is the most relevant creep mechanism for long-term geophysical modeling. Lastly, tertiary (accelerating) creep, which sometimes is not observed in experiments, is the last creep regime in which the strain rate increases exponentially with time.
Secondary creep is often described by a power law, which relates the strain rate to the differential stress and temperature (Hirth and D. Kohlstedt 2004):

$$\dot{\epsilon} = A\sigma^n d^{-m} fH_2O^r e^{\alpha\phi} e^{-\frac{Q+PV}{RT}} \tag{5}$$

where $\dot{\epsilon}$ is the strain rate, A is a material constant, σ is the differential stress, n is the stress exponent, d is the grain size with the exponent m, Q is the activation energy, R is the universal gas constant and T is the temperature. The power law describes the relationship between the strain rate and the differential stress, where the stress exponent n determines how sensitive the strain rate is to changes in the differential stress. The activation energy Q describes how sensitive the strain rate is to changes in temperature. The work done against the hydrostatic confining pressure P to create or move a defect (where V is the activation volume) also contributes to the energy required for deformation. For lithospheric-scale geodynamics, where pressures are relatively low compared to the deep mantle, the term PV is sometimes neglected. The term fH_2O^r accounts for the effect of water on the creep behaviour, where fH_2O is the water fugacity and r is the water content. In geophysical modeles, this term is often neglected, as it is already implicitly accounted for in the other experimentally derived parameters, i.e. A. The term $e^{\alpha\phi}$ accounts for the effect of porosity on the creep behaviour, where α is a material constant and ϕ is the porosity. At high stresses, for example in the lower crust or lithosphere, the applied stress is much larger than the threshold stress, and the term becomes negligible as well. Therefore the equation simplifies to:

$$\dot{\epsilon} = A\sigma^n d^{-m} e^{-\frac{Q}{RT}} \tag{6}$$

Solving this for the differential stress σ yields the following equation:

$$\sigma = \left(\frac{\dot{\epsilon} d^m}{A}\right)^{\frac{1}{n}} e^{\frac{Q}{nRT}} \tag{7}$$

1.2 Diffusion Creep

Diffusion creep characterizes the deformation of rocks by diffusion of vacancies through the crystal lattice of minerals. The main principle that governs a crystal lattice structure is Gibbs free energy. It is the energy that is available in a closed system to perform work. Naturally, every system strives for a state of equilibrium, which is the state of lowest Gibbs free energy. It is defined as:

$$\Delta G = \Delta H - T\Delta S \tag{8}$$

where G is the Gibbs free energy, H is the enthalpy, T is the temperature and S is the entropy. Enthalpy is a measurement of the total energy of a system, while entropy describes the degree of disorder within a system. At temperatures above $0K$, atoms within a crystal lattice are no longer static, but vibrate around their equilibrium position due to thermal energy. Therefore it has nonzero enthalpy. Introducing defects within a crystal lattice, such as vacancies or self-interstitials, has two effects on Gibbs free energy. First, in increases enthalpy, as creating defects within the lattice costs energy, since it requires breaking atomic bonds and rearranging atoms, which leads to an increase in enthalpy. Second, it increases entropy as well, since defects increase the number of possible arrangements of atoms within the lattice, which leads to an increase in disorder. Entropy therefore can be written as:

$$S = k_B \log \frac{N!}{(N-n)!n!} \tag{9}$$

where k_B is the Boltzmann constant, N is the total number of atoms in the lattice, and n is the number of defects. Entropy increases nonlinearly with temperature, while enthalpy increases linearly with defect concentration. As a result, the Gibbs free energy is minimized at a finite defect concentration, giving an equilibrium state.

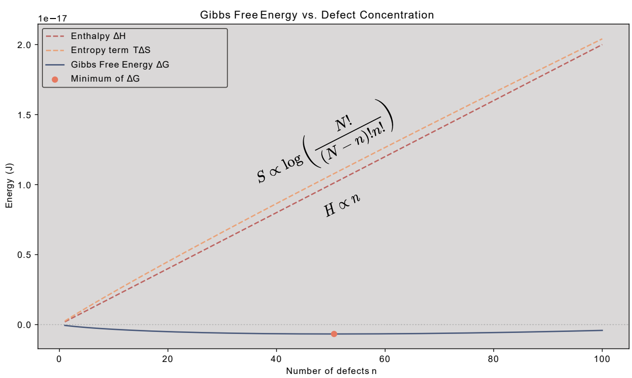
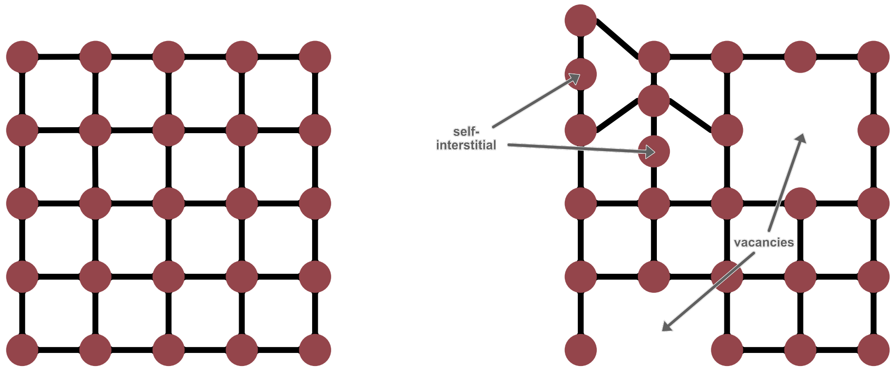


Figure 1. Gibbs free energy as a function of defect concentration. The minimum of the curve represents the state of equilibrium, where the system has a nonzero amount of defects in the crystal lattice. The increase in Gibbs free energy is due to the increase in enthalpy, while the decrease in Gibbs free energy is due to the increase in entropy.



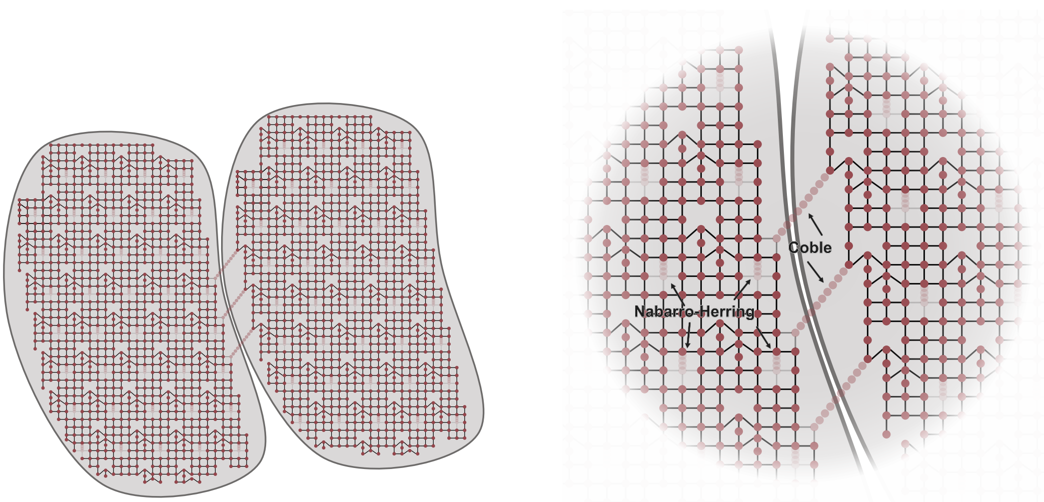
(a) Perfect crystal lattice at $T = 0K$

(b) Crystal lattice in equilibrium at $T > 0K$

Figure 2. Crystal lattice structure at different temperatures. At $T = 0K$, the crystal lattice is perfect, while at $T > 0K$ the atoms vibrate around their equilibrium position, leading to a nonzero amount of defects in the crystal lattice.

Diffusion creep can be either through the grains interiors (Nabarro-Herring creep) or along grain boundaries (Coble creep) (Ranalli 1995). The stress exponent n for the constitutive flow law is equal to 1. Hence the final equation for the differential stress for diffusion creep is:

$$\Delta\sigma_{diff} = \left(\frac{\dot{\epsilon}\, d^m}{A}\right) e^{\frac{Q}{RT}} \tag{10}$$



(a) Simplified crystal lattice in a two-grain system

(b) Closeup of the grain boundary

Figure 3. Different diffusion mechanisms by the propagation of vacancies along the crystal lattice: (b) shows Nabarro-Herring creep, where vacancies diffuse through grain interiors, and Coble creep, where vacancies diffuse along grain boundaries.

1.3 Dislocation Creep

Dislocations within a lattice are line defects, which are caused by the misalignment of atoms within the lattice structure. Their motion by gliding or climbing is essential for deformation and fullfils a similar role as point defects in diffusion creep. These dislocations are usually not inherent in a lattice in equilibrium, since the change of entropy forming from a formation of vacancies constrained to a line is significantly lower than the change of entropy forming from randomly distributed vacancies in a crystal, as is apparent from Eq. 9. Therefore dislocations are usually introduced by external forces, such as stress or strain, which cause the atoms to misalign and form dislocations. Dislocations are defined by two vectors: the Burgers vector **b**, which describes the magnitude and direction of the lattice distortion caused by the dislocation, and the line vector **l**, which describes the direction of the dislocation line. The Burgers vector is defined as the difference between two lattice vectors in a perfect crystal lattice and is usually perpendicular to the dislocation line. Depending on their alignment, dislocations can either be edge dislocations (perpendicular) or screw dislocations (parallel).

Dislocation creep involves the movement of dislocations through the crystal lattice which causes plastic deformation of the individual crystals, and thus the material itself. This deformation mechanism is independent of grain size and dominates in the lithospheric mantle, lower continental crust (where T allows), early in the development of a shear zone, and in hotter-than-average mantle (e.g., plumes and convection cells, due to lower viscosity). For dislocation creep, the stress exponent n is usually between 3 and 5. Furthermore it takes the form of Harper-Dorn creep (dislocation glide) as soon as the Peierls stress of a crystal is reached (Wang and Nieh 1995). Peierls stress is associated with the forces needed to move a dislocation within a plane of atoms in the unit cell. The mechanism involves the climb of dislocations in a material of constant, low dislocation density and large grain-size, so that grain boundary diffusion does not play a major role. This mechanism becomes important at high strain rates (high differential stress). Dislocation glide can be considered a linearized or approximate form of the creep law (Cacace and Scheck-Wenderoth 2016). This kind of creep can be described by:

$$\dot{\epsilon} = \dot{\epsilon}_{Do} \cdot e^{-\frac{Q}{RT}(1-\frac{\sigma_d}{\sigma_{Do}})^p} \tag{11}$$

with σ_0 being the Peierls critical stress, ϵ_0 the Dorn critical strain rate and Q_{Dorn} the activation energy for dislocation glide.

1.4 Low Temperature Plasticity (Peierls Creep)

Low temperature plasticity (LTP) is a deformation mechanism that occurs at low temperatures and high pressures, and is a form of dislocation glide. It is characterized by the movement of dislocations through the crystal lattice, which causes plastic deformation of the material. It is dependent on the Peierls stress, which is the stress required to move a dislocation within a plane of atoms, which is significantly lower than the stress required for dislocation creep or diffusion creep at lower temperatures. Therefore it can be observed within unusually cold lower crust or upper lithosphere.

At stresses higher than a critical threshold (commonly 200 MPa, the Peierls critical stress threshold), deformation by dislocation glide supersedes standard dislocation creep. The equation for the combined mechanism reflects this transition:

$$\Delta\sigma_{creep} = \begin{cases} \left(\frac{\dot{\epsilon}}{A}\right)^{\frac{1}{n}} e^{\left(\frac{Q}{nRT}\right)} & \text{if } \Delta\sigma_{creep} < 200\text{MPa} \\ \sigma_{Peierls}\left(1 - \left[-\frac{RT}{Q_{Peierls}} \ln\left(\frac{\dot{\epsilon}}{\epsilon_0}\right)\right]^{\frac{1}{2}}\right) & \text{if } \Delta\sigma_{creep} > 200\text{MPa} \end{cases} \tag{12}$$

To ensure continuity across the transition from dislocation creep to Peierls creep, the activation energy for Peierls creep ($Q_{Peierls}$) is dynamically evaluated. It is constrained such that the predicted stresses and strain rates perfectly match at the transition point, providing a mathematically robust and physically consistent rheological profile across varying depths.

2. Methods and Algorithm

2.1 From the Lab to Numerical Models

Laboratory experiments are typically performed on single minerals under controlled conditions, such as constant temperature, pressure, and strain rate, to determine the onset of deformation. The results from these experiments are used to derive constitutive flow laws, which describe the relationship between stress and strain rate. These laws are often published in terms of parameters - such as A, n, m, and Q - and are used in equation 6. However, applying these laboratory-derived parameters to realistic geophysical models requires significant extrapolation beyond experimental conditions. Furthermore, there are a couple more important aspects one has to be cautious about. First, A is not a material constant, but a property function $A(P, T, b, fO_2, ...)$, which is dependent on several parameters, such as pressure temperature, Burgers vector, oxygen fugacity and many more. Usually a set of parameters are therfore already integrated into the value of A, which is different from publication to publication. Hence, it is important to modify certain published values in order to make them consistent with equation 6. Second is the upscaling needed from the strain rates of the laboratory to those of the lithospere in nature. Lastly, laboratory experiments are often performed with a uniaxial setup, where the sample is deformed under a constant load in one direction, while the other two directions are constant. Mathematically, the stress and strain rate can be described by second order tensors:

$$\boldsymbol{\sigma}'_{ij} = \begin{pmatrix} \sigma'_{xx} & 0 & 0 \\ 0 & -\frac{1}{2}\sigma'_{xx} & 0 \\ 0 & 0 & -\frac{1}{2}\sigma'_{xx} \end{pmatrix}, \quad \dot{\boldsymbol{\epsilon}}'_{ij} = \begin{pmatrix} \dot{\epsilon}'_{xx} & 0 & 0 \\ 0 & -\frac{1}{2}\dot{\epsilon}'_{xx} & 0 \\ 0 & 0 & -\frac{1}{2}\dot{\epsilon}'_{xx} \end{pmatrix} \tag{13}$$

where σ_{ij} is the deviatoric stress tensor and $\dot{\epsilon}_{ij}$ is the strain rate tensor. In nature, however, the stress and strain rate tensors are usually not diagonal, but have non-zero off-diagonal components, which represent shear stresses and shear strain rates. Therefore we need to parameterize equation 6 as an expression of effective viscosity η_{eff} (Gerya 2019), which is defined as the ratio of the second invariants of the deviatoric stress (σ_{II}) and strain rate ($\dot{\epsilon}_{\mathrm{II}}$).

The relationships for σ_{II} and $\dot{\epsilon}_{\mathrm{II}}$, which are a scalar measures of the stress and strain rate that is invariant under coordinate transformations are given as:

$$\sigma_{\mathrm{II}} = \frac{1}{\sqrt{3}}\sigma_d \quad \text{and} \quad \dot{\epsilon}_{\mathrm{II}} = \frac{\sqrt{3}}{2}\dot{\epsilon} \tag{14}$$

Combining equations 7 and 19 with the relationships in equation 14 yields the effective viscosity as a function of the strain rate:

$$\eta = F\frac{1}{(A)^{\frac{1}{n}}(d)^{-\frac{m}{n}}(\dot{\epsilon}_{\mathrm{II}})^{\frac{n-1}{n}}}e^{\frac{Q}{nRT}} \tag{15}$$

with F being a conversion factor needed for geophysical modeling when using laboratory-derived values from uniaxial compressed rock samples (Ranalli 1995). It is defined as:

$$F = \frac{1}{(2)^{\frac{n-1}{n}}(3)^{\frac{n+1}{2n}}} \tag{16}$$

In case of the experiment would be a simple shear setup, rather than uniaxial compression, the factor F would change according to the relationship between the second invariant of the deviatoric stress and strain rate and the shear stress and strain rate, respectively. In that case, the factor F would be defined as:

$$F = \frac{1}{(2)^{\frac{2n-1}{n}}} \tag{17}$$

While this factor is usually applicable to both dislocation and diffusion creep, the rather isotropic nature of diffusional processes (which results in a factor of 3 being added to the equation) as well as $n = 1$ for diffusion creep leads to this factor being effectively 1 for diffusion creep. Therefore, the effective viscosity for diffusion creep can be expressed as:

$$\eta_{diff} = \frac{1}{Ad^{-m}} e^{\frac{Q}{RT}}, \quad \eta_{disl} = F \frac{1}{(A)^{\frac{1}{n}} (\dot{\epsilon}_{\mathrm{II}})^{\frac{n-1}{n}}} e^{\frac{Q}{nRT}} \tag{18}$$

To ensure coordinate invariance and frame indifference, we formulate the effective viscosity using the second invariants of the stress (σ_{II}) and strain rate ($\dot{\epsilon}_{\mathrm{II}}$) tensors:

$$\eta_{eff} = \frac{\sigma_{\mathrm{II}}}{2\dot{\epsilon}_{\mathrm{II}}} \tag{19}$$

where the factor 2 is used to account for the rotational invariants of the system.

Furthermore, rocks experiencing stress do deform with a diffusional and dislocational creep mechanism simultaneously. Assuming constant applied deviatoric stress, the deviatoric strain rate consists of equal contributions from both mechanisms, which can be expressed as:

$$\sigma_{ij} = \sigma'^{diff}_{ij} + \sigma'^{disl}_{ij} \tag{20}$$

where $\dot{\epsilon}'^{diff}_{ij}$ is the deviatoric strain rate due to diffusion creep and $\dot{\epsilon}'^{disl}_{ij}$ is the deviatoric strain rate due to dislocation creep. The total effective viscosity can then be expressed as:

$$\frac{1}{\eta_{eff}} = \frac{1}{\eta_{diff}} + \frac{1}{\eta_{disl}} \tag{21}$$

2.2 Strength Profiles

In geomechanics an important concept is the strength of a material and is related to the stress (loading) a sample of rock can bear before permanent deformation occurs. It is differentiated between yield strength, which is the stress level at which a material starts to deform plastically, and failure strength, which is the maximum stress a material can withstand before failure or fracturing. It has been long recognized that the mode of lithospheric rock deformation changes with depth in the plate. Usually it is assumed that rocks at shallow crustal levels deform in a brittle manner, while rocks at deeper crustal and mantle levels in a rather ductile manner. Assuming a (quasi)-static rheological response of lithospheric rocks to loading, the concept of yield strength envelopes (YSE) (Evans and Goetze 1979; Brace and D. L. Kohlstedt 1980) combines brittle and ductile deformation laws with observations indicating that rocks deform by the mechanism requiring the least differential stress at a given depth. Thus, the depth-dependent change of maximum strength of a rock is provided by the envelope that results from assessing the differential stress that is the least among those associated with the different deformation mechanisms. The calculation of a YSE thus follows:

$$\Delta\sigma_Y = \min(\Delta\sigma_B, \Delta\sigma_{creep}, \Delta\sigma_{peierls}) \tag{22}$$

where $\Delta\sigma_Y$ is the yield stress, $\Delta\sigma_B$ is the Byerlee stress, $\Delta\sigma_{creep}$ is the combined viscous creep stress (incorporating both diffusion and dislocation creep via the harmonic mean of their effective viscosities), and $\Delta\sigma_{peierls}$ is the Peierls stress for low-temperature plasticity. These stresses are calculated using their respective equations. For brittle failure Eq. 4 is used. Viscous creep is evaluated continuously (Eq. 21), rather than switching abruptly between dislocation and diffusion creep mechanisms.

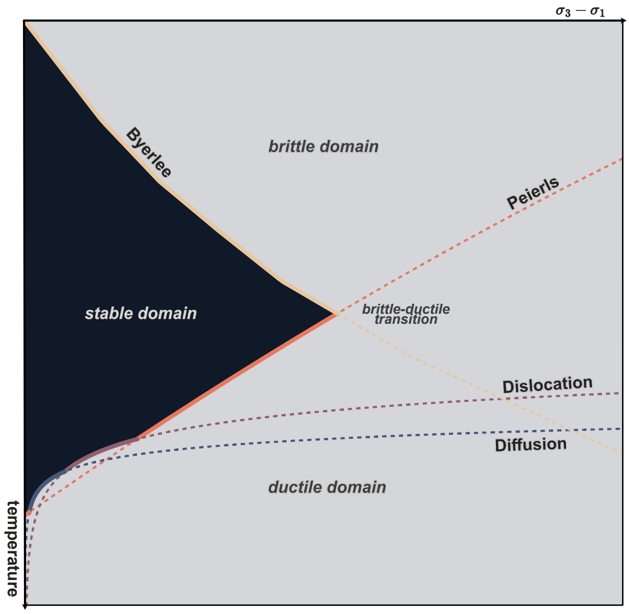


Figure 4. Example yield strength envelope showing the temperature-dependent transition between brittle and ductile deformation mechanisms. Since Byerlee's law is a function of confining pressure, a geothermal gradient is assumed. This gradient is not strictly linear with depths, which is why the brittle part of the YSE is also not linear.

Common to both rheological descriptions is that the shape of the strength profile determines at which depth differential elastic stresses can accumulate. It therefore imposes self-consistent bounds to the amount of energy that can be stored and possibly released whether seismically or aseismically. The brittle-ductile transition (BDT) at a given strain rate is given by the depth at which the two curves intersect (Scholz 1988). However, whether a rock behaves in a rather brittle or ductile fashion depends on specific environmental conditions as: the level of confining pressure; the presence and type of fluids; temperature and strain rate boundary conditions; and the total strain exerted on a rock.

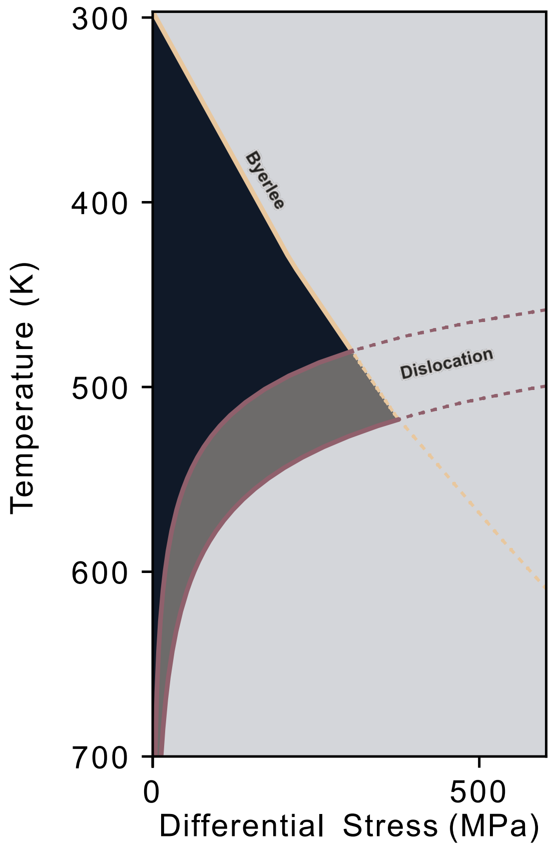
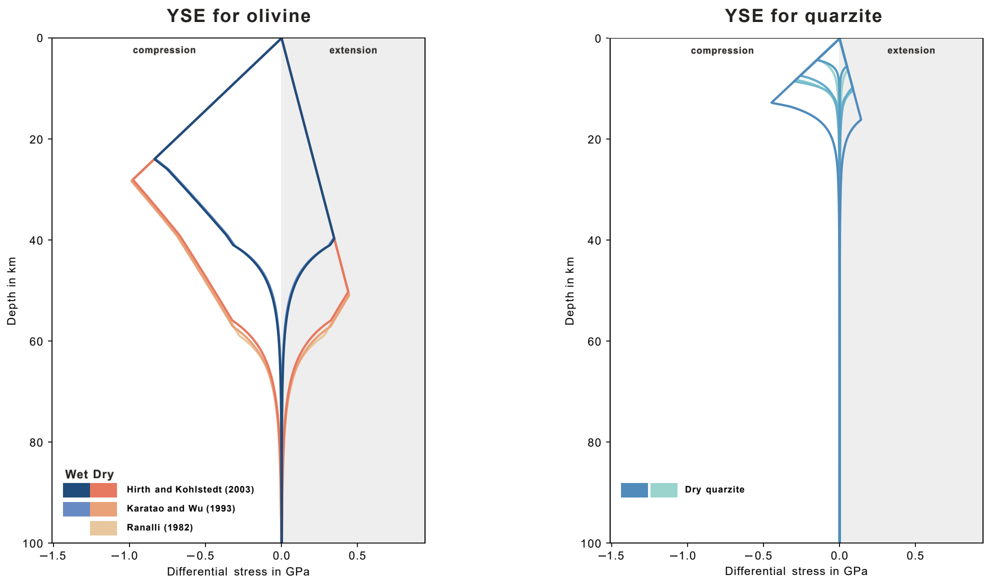


Figure 5. Influence of the laboratory configuration on the yield strength profile. If not accounted for the uniaxial experimental setup, the strength profile may be significantly overestimated.

Figure 5 shows the influence of the laboratory configuration on the yield strength profile. If not accounted for the uniaxial experimental setup, the strength profile may be significantly overestimated. This is because the laboratory experiments do not capture the full stress and strain rate tensors in terms of 3D distribution, but rather only the diagonal components of theses tensors. Therefore the yield strength profiles without correction are a factor of F times larger, as shown in equation 16, than they would be in a realistic geophysical model. Therefore it is important to account for this factor when calculating the yield strength profile from laboratory experiments.

2.3 RheologyExplorer

Cross-plotting yield strength profiles for different single mineral compositions enables a direct comparison of their mechanical behavior under identical pressure and temperature conditions. Therefore it reveals how variations in intrinsic parameters, such as grain size, activation energy, and stress exponent - which are dependent on the experimental setup - affect the yield strength and deformation mechanisms of each mineral. This approach highlights the diversity in strength characteristics among minerals and aids in interpreting lithospheric strength profiles for geophysical modeling.



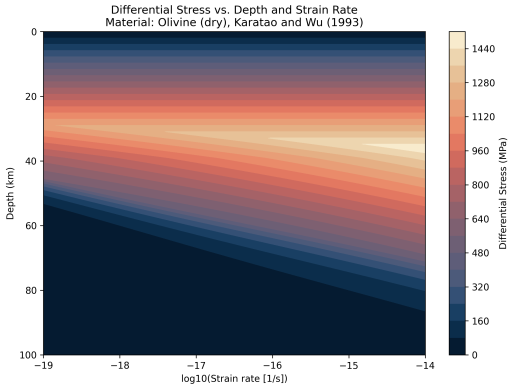
(a) Yield strength profile for olivine.

(b) Yield strength profile for quartzite.

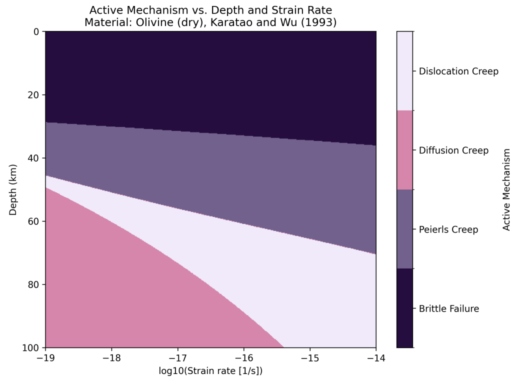
Figure 6. Comparison of yield strength profiles for olivine and quartzite under same pressure and temperature conditions. (a) shows olivine, (b) shows quartzite. The yield strength profiles are calculated using the geothermal gradient from (McKenzie, Jackson, and Priestley 2005).

As can be seen in figure 6, the yield strength profiles for olivine are almost the same across different literature values, whereas the profiles for quartzite are very different. This shows the differences between different experimental setups. Under wet condition, the rock's strength is significantly lower than under dry conditions, which is due to the presence of water in the pore space of the rock.

Since a rock's strength is not only dependent on its mineralogical properties, but also a function of strain rate and temperature at the same time, the coupling between layers may vary drastically, based on the models assumptions (Burov and Diament 1995).



(a) Strength of olivine as a function of temperature and strain rate.



(b) Respective mechanisms of deformation.

Figure 7. The strength of olivine from Karato and Wu (1993) as a function of temperature and strain rate. The corresponding deformation mechanisms are shown in (b).

As stated before, Byerlee's law is a function of confining pressure and therefore pore fluid pressure. The pore fluid pressure is often assumed to be hydrostatic, meaning it increases linearly with depth according to the specific weight of the fluid column. However, this assumption is not always valid, as the pore fluid pressure can be influenced by tectonic processes, such as faulting or folding, which can lead to overpressure or underpressure conditions (Hubbert and Rubey 1959). Therefore it is important to consider the pore fluid pressure when calculating the yield strength profiles, as it can significantly affect the results.

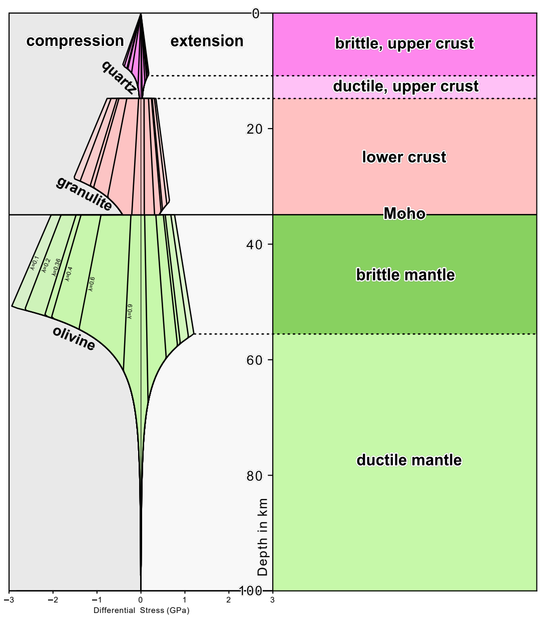


Figure 8. Strength Profiles with different pore fluid pressure conditions.

Figure 8 shows the effect of different pore fluid pressure conditions on the yield strength. It only affects the linear Byerlee stress, while diffusion and dislocation creep remain unaffected. An increase in pore fluid pressure, and therefore λ decreases the yield strength, making the rock effectively weaker. In stress computations, a value of $\lambda = 0.36$ is often used, which represents the hydrostatic pore fluid pressure in the Earth's crust. It comes from the simple assumption of water having a density of $1000\,\frac{kg}{m^3}$ while the sourounding rock has a density of $2760\,\frac{kg}{m^3}$, which is a common value for the Earth's crust. Therefore:

$$\lambda = \frac{\rho_{water}}{\rho_{rock}} = \frac{1000\,\frac{kg}{m^3}}{2760\,\frac{kg}{m^3}} = 0.36 \tag{23}$$

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